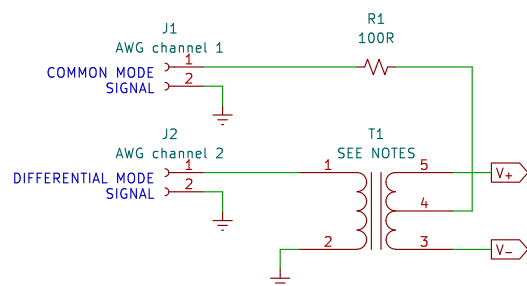


TEST RIG

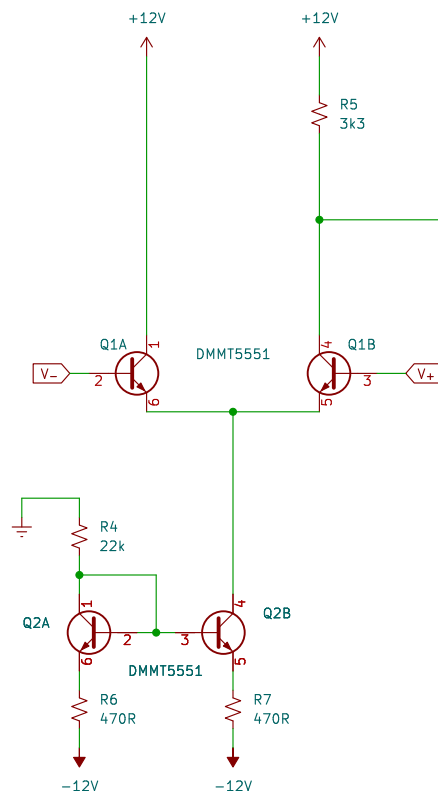


T1 is a small 120V:6.3VCT power transformer Triad Magnetics F-14X or similar. Suitable transformers can often be scavenged from discarded equipment with linear power supplies.

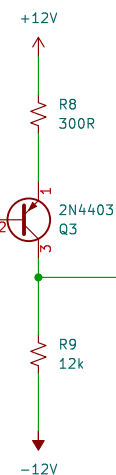
Test rig allows testing the pair's response to common mode and differential signals, together or separately.

When applying the common mode signal by itself, it is advantageous to short pins 1 and 2 of Q1 with a short jumper to avoid picking up spurious differences.

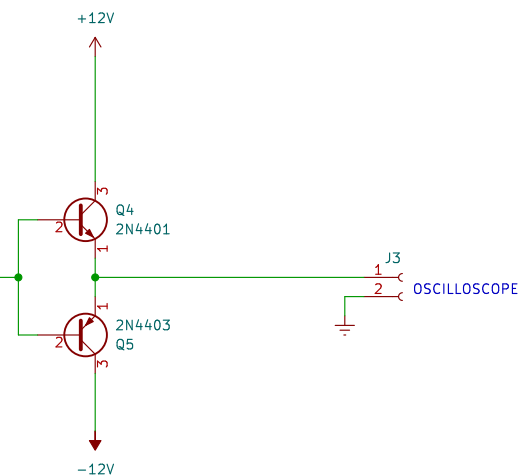
LONG-TAILED PAIR



GAIN STAGE



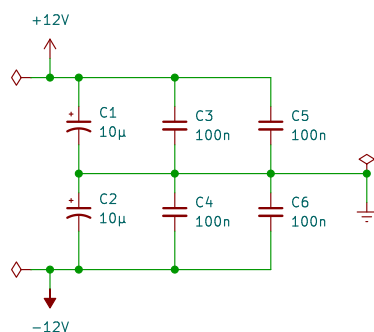
FOLLOWER STAGE



This circuit can be breadboarded starting with the circuit from episode 15, modifying R5 from 24k to 3k3, and incrementally adding the gain stage and follower stage.

1. Replace R5, add Q3, R8, R9, C4 and C6; observe common-mode and differential gains at Q3's collector.
2. Add Q4 and Q5. Observe crossover distortion at the output.

POWER AND BYPASSING



Place C3, C4 as close to Q1 as possible.
Place C5, C6 as close to Q4 and Q5 as possible.

500 μ A CURRENT SOURCE



Transistors 101, episode 16

The differential pair: getting more gain

Kludges from Kevin's Cave

Sheet: /

File: LongTailedPair3.kicad_sch

Title: Long-tailed pair: the basic circuit

Size: USLetter Date: 2026-05-13

KiCad E.D.A. kicad 7.0.11+dfsg-1build4

Rev: B

Id: 1/1